1. List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

IP Class	First Octet Range	Default Subnet Mask	Purpose	Real-World Example
Class A	1 – 126 *(0 and 127 are reserved)*	255.0.0.0 (/8)	Large enterprise or government networks with millions of hosts	Large multinational corporation core network
Class B	128 – 191	255.255.0.0 (/16)	Medium-to-large networks (universities, mid-sized companies)	University campus network
Class C	192 – 223	255.255.255.0 (/24)	Small local area networks (LANs) and small businesses	Small office Wi-Fi router or home network
Class D	224 – 239	None (Not applicable)	Multicast communication (one-to-many data streams)	IPTV live video streaming broadcast group
Class E	240 – 255	None (Not applicable)	Experimental, research, and military development	R&D testing in a closed laboratory network

2. Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.

Hint: Show your working steps for each calculation.

To find the network address, broadcast address, and host range, we can look at the IP and subnet mask in terms of their octets (the four blocks of numbers).

Because the subnet mask is 255.255.255.0, it tells us exactly where the "Network" part ends and the "Host" part begins. Every position with a 255 is locked by the network, and the position with a 0 belongs to the hosts.

Here are the step-by-step calculations:

1. Network Address

The network address represents the entire street or neighbourhood. To find it, you **keep the network parts exactly the same** and **change all host parts to 0**.

- **Step 1:** Look at the mask 255.255.255.0. The first three blocks are locked (255), and the last block is free (0).
- **Step 2:** Copy the first three blocks from the IP address: 192.168.10.
- **Step 3:** Change the last block (the host part) to 0.
- **Result: 192.168.10.0**

2. Broadcast Address

The broadcast address is used to send a message to every device on this specific network at the same time. To find it, you **keep the network parts the same** and **change all host parts to the maximum value, 255**.

- **Step 1:** Copy the locked network blocks: 192.168.10.
- **Step 2:** Set the last block (the host part) to its maximum possible number, 255.
- **Result: 192.168.10.255**

3. Range of Valid Host Addresses

The usable host range consists of all the numbers sitting safely **between** the Network Address and the Broadcast Address. You cannot use the .0 or .255 numbers for actual devices (like computers or phones) because they have special jobs.

- **Step 1 (First Usable Host):** Add 1 to the last block of your network address (192.168.10.0 + 1 = 192.168.10.1).
- **Step 2 (Last Usable Host):** Subtract 1 from the last block of your broadcast address (192.168.10.255 - 1 = 192.168.10.254).
- **Result: 192.168.10.1 to 192.168.10.254** (giving you a total of 254 usable addresses).

3. Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

To create this plan for your food delivery startup, we will take the original **10.0.0.0/24** network and slice it into **4 equal pieces**.
Think of the **10.0.0.0/24** network as a large piece of land with **256 total addresses** (from .0 to .255). To create 4 equal neighborhoods, we divide 256 by 4, giving each subnet exactly **64 addresses**.
To do this, we change the subnet mask from /24 to **/26** (which looks like **255.255.255.192** in decimal format).
Here is the complete subnetting plan for your startup:

Zomato-style Subnetting Plan

Subnet Group	Network Address	Subnet Mask (Decimal)	Subnet Mask (CIDR)	Usable Host IP Range	Broadcast Address
Delivery Partners	10.0.0.0	255.255.255.192	/26	10.0.0.1 to 10.0.0.62	10.0.0.63
Restaurants	10.0.0.64	255.255.255.192	/26	10.0.0.65 to 10.0.0.126	10.0.0.127
Customers	10.0.0.128	255.255.255.192	/26	10.0.0.129 to 10.0.0.190	10.0.0.191
Admins	10.0.0.192	255.255.255.192	/26	10.0.0.193 to 10.0.0.254	10.0.0.255

4. You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.

Constraint: Show your calculation steps and final answers.

To create at least 20 subnets from the **172.16.0.0/16** network, we need to borrow bits from the host part to make enough slices.

Here are the calculation steps broken down simply:

Step 1: Find how many bits to borrow

Computers create subnets using powers of 2 (binary). We need to find how many bits to borrow to get *at least* 20 subnets.

- 1 bit = 2^1 = 2 subnets
- 2 bits = 2^2= 4 subnets
- 3 bits = 2^3 = 8 subnets
- 4 bits = 2^4 = 16 subnets (Not enough, we need 20)
- **5 bits = 2^5 = 32 subnets** (This works! This gives us 32 subnets, which covers our need for 20).

We must borrow **5 bits**.

Step 2: Calculate the new Subnet Mask

The original network mask is **/16**. We add the 5 borrowed bits to it:
New Mask = 16+5= /21

To turn /21 into a human-friendly decimal mask, we look at the 4 blocks (octets).

- The first two blocks are fully locked: 255.255
- The third block gets the 5 borrowed bits. The 8 binary slots in an octet are worth: 128, 64, 32, 16, 8, 4, 2, 1.
- Adding the first 5 slots together: (128 + 64 + 32 + 16 + 8 = 248).
- The last block is left untouched: 0.

Final Subnet Mask: 255.255.248.0 (or **/21**)

Step 3: Calculate the number of hosts per subnet

An IP address has 32 bits total. Since our new mask uses 21 bits for the network, we subtract that to see how many bits are left for device IP addresses (hosts):
Remaining Host Bits=32-21=11 bits)

Now, we calculate how many total IP numbers 11 bits can create:
Total IPs}=2^{11}=2048 addresses)

Just like your home network, we must subtract **2 addresses** from every subnet (1 for the *Street Name / Network Address* and 1 for the *Megaphone / Broadcast Address*):
Usable Hosts =2048-2=2046

Final Answers

- **New Subnet Mask: 255.255.248.0** (CIDR: **/21**)
- **Available Hosts per Subnet: 2,046 usable devices** per building.

5. A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

Here is a short, friendly message you can send back to your friend to explain the problem:

"Hey! The reason your device isn't connecting is because of a typo in the IP address: 192.168.1.300 is actually an impossible IP address.

Here is why:

- **The 255 Limit:** IP addresses are split into four number blocks, and no single block can ever go higher than 255.
- **The Error:** Your last block is 300, which is too high for a computer to understand. It's like trying to write a calendar date as October 45th!

What a valid IP looks like:

Assuming your home network uses the standard default mask (255.255.255.0), the last number has to sit safely between 1 and 254.

A correct, working IP address for your device should look something like 192.168.1.30 or 192.168.1.45. Check your settings and make sure that last number didn't accidentally get an extra zero added to it!"